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Quinidine Drug Resistance transporter Knockout *Candida* cells modulate glucose transporter expression and accumulate metabolites leading to enhanced azole drug resistance

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Abstract

ATP-binding cassette (ABC) and Major Facilitator Superfamily (MFS) transporters have been known to play an important role in the development of multidrug resistance (MDR) in various fungal species. Although the significance of ABC transporters in MDR evolution is well established, the MFS exporters, on the other hand, have received little attention. This study highlights the role of *QDR* (quinidine drug resistance) transporters (*CaQDR1*, *CaQDR2*, and *CaQDR3*), a subfamily of MFS, in imparting pathogenicity and virulence to *Candida albicans*. Here we report that the transcriptomic analysis of *QDR* knockout (*QDRKO*) strains compared to the wild-type (WT) strain of *C. albicans* shows differential expression of some essential virulence-associated categories of genes. The chitin and β -glucan synthases, mannosyl transferases, vacuolar, ion transporters, acid phosphatase, and different sugar transporter (*HGT8* and *HGT9*) encoding genes. Although some of the phenotypic assays could not show any considerable differences in the growth of knockout strains under relevant stresses, however, we found that the *QDRKO* strains show elevated expression levels of different *HGT* genes, particularly under glucose limiting conditions which was also reflected by the higher intracellular glucose accumulation levels. All the strains (*QDRKO* and WT) followed a similar pattern in the accumulation of metabolite glycerol. Interestingly, *QDRKO* strains exhibit an enhanced azole resistance than the parental *Candida* strain, particularly at a low glucose concentration of the culture media. Our findings suggest that the deletion of *QDR* genes (individually or collectively) alters cellular pathways, particularly those associated with glucose and glycerol accumulation. This possibly provides the cells with a mechanism to overcome stress and partially maintain the cellular pathogenicity/virulence in the absence of *QDR* MFS transporters.

Keywords: *Candida albicans*; quinidine drug resistance (QDR) transporters; glucose accumulation; glucose transporters; pathogenicity.

Running title: Systematic interplay between *C. albicans* membrane MFS-transporters

Abbreviations

ABC: ATP-binding cassette; **MFS:** Major Facilitator Superfamily; **MDR:** Multidrug resistance; **HIV:** Human immunodeficiency virus; **YEPD:** Yeast extract peptone-dextrose; **PBS:** Phosphate buffered saline; **cdNA:** Complementary DNA; **PCR:** Polymerase chain reaction; **RT-PCR:** Reverse transcription– Polymerase chain reaction; **qRT-PCR:** Reverse quantitative transcription–Polymerase chain reaction; **EDXRF:** Energy Dispersive X-ray Fluorescence Technique

1. Introduction

The organisms like bacteria, fungi, viruses, and parasites represent the most common infection-causing agents which can employ different MDR strategies to combat the effect of various antimicrobials. Today MDR represents a significant global public health threat because of the increasing deaths caused by fatal diseases, like tuberculosis, malaria, pneumonia, HIV, yeast infections, and other related lethal diseases (Tanwar et al., 2014). For example, it has been reported that around 90% of the fungal-associated deaths occur as a result of the species belonging to *Cryptococcus*, *Candida*, *Aspergillus*, and *Pneumocystis* genera (Brown et al., 2012).

The frequency of fungal infections caused by the human pathogenic fungi has been increasing drastically, possibly due to the increase in the number of immunocompromised individuals generally affected with HIV, asthma, cancer chemotherapy, organ, bone marrow transplantation surgeries, and corticosteroid therapies (Bongomin et al., 2017). The drastic rise in the incidence of fungal infections has eventually led to the development of different antifungal agents for the necessary treatment and prevention. However, since only a few classes of antifungals are available, the appropriate treatment options become limited (Kathiravan et al., 2012; Cowen et al., 2015).

Candida species is the most prevalent cause of human systematic fungal infections, posing a high risk of morbidity and mortality. *C. albicans*, a commensal organism, colonizing mostly the urogenital and gastrointestinal tracts, mucosa and skin, causing superficial oral and vaginal infections like oral candidiasis and vaginitis, dermal/skin infections, as well as serious systematic fungal infections, is a major human fungal pathogen, which can thrive in diverse niches (Calderone et al., 2002; Brown et al., 2006; Rodaki et al., 2009; Kühbacher et al., 2017; Cortegiani et al., 2018). Although *C. albicans* is the primary agent of nosocomial fungal infections among *Candida* species, several non-albican species viz; *C. glabrata*, *C. krusei*, *C. parapsilosis*, and *C. tropicalis*, cause invasive infections with increasing rates (Prasad et al., 2014). Although multiple treatments/therapies are currently being employed to reduce the burden of increasing human fungal infections, only moderate success has been achieved (Redhu et al., 2018). Furthermore, extended use of antifungal agents, particularly azoles, has resulted in the development of drug resistance, and there is the progression of infection despite appropriate antifungal therapy (Shah et al., 2015). Moreover, these fungal pathogens have gained resistance by developing various strategies like target mutation/alteration: ergosterol biosynthetic pathway alterations (point mutations in *ERG11*); target gene overexpression: overexpression of *ERG11* either through duplication (gene dosage effect) of the gene or by upregulation (by *trans*-acting factors) of the gene; overexpression of the drug efflux pump-encoding genes *CDR1* (*Candida* drug resistance 1), *CDR2* (*Candida* drug resistance 2), belonging to the ABC (ATP-binding cassette) superfamily and *MDR1*, belonging to the MFS (major facilitator superfamily) transporters; and utilization of other compensatory pathways to combat the effect of currently available antifungal drugs (Vandeputte et al., 2012; Shah et al., 2014; K. Redhu et al., 2016).

In *C. albicans*, drug transporters play an essential role in the development of MDR and the major transporters responsible are *Cdr1p* and *Cdr2p* (ABC proteins) and *Mdr1p* (MFS proteins) (Shah et al., 2014; Bhattacharya et al., 2016). Of late, a new study has confirmed the role of a plasma membrane-localized transporter (CDR6/ROA1) in effluxing various xenobiotic types like Berberine, Paraquat, and Rhodamine 123 (Khandelwal et al., 2018). The overexpression of the efflux transporter genes aids in the rapid efflux of incoming drugs prevent cells from accumulating lethal drug doses, enabling survival and hence resistance (Ernst et al., 2010; Shah

et al., 2014; K. Redhu et al., 2016; Prasad et al., 2016). Another important mode of resistance adopted by *Candida* species is the formation of complex drug-resistant biofilms. Among the *Candida* species, the biofilms formed by *C.albicans* are highly extensive and complex, which adds to its pathogenicity. Biofilms are believed to be resistant because of various factors, including drug-tolerant persister cells (Gulati et al., 2016). Due to multiple drug resistance (MDR) mechanisms, these fungal pathogens survive high doses of antifungal drugs.

In the *S. cerevisiae* genome, a total of 22 MFS drug transporters have been reported which generally belongs to two families: the 12-spanner drug: H⁺ antiporter family 1 (DHA1), and the 14-spanner drug: H⁺ antiporter family 2 (DHA2) having 12 and 14 transmembrane segments (TMS) respectively (Costa et al., 2014). The expression of MFS-MDR transporters in *S. cerevisiae* has been shown to provide resistance to various chemical stresses, generally ranging from anticancer, antifungal, and antimalarial drugs to herbicides (Sá-Correia et al., 2009; Mira et al., 2010). The *S. cerevisiae* DHA1 sub-family includes the *Qdr1*, *Qdr2*, and *Qdr3* transporters. *QDR1* (quinidine resistance 1), *QDR2*, and *QDR3* were found to be involved in resistance against the drug quinidine, the herbicide BBN (Barban), anti-cancer compounds, and several commonly used antifungal agents like ketoconazole and fluconazole (Nunes et al., 2001; Vargas et al., 2004; Tenreiro et al., 2005). Based on the computational analysis, 95 putative MFS transporters have been reported in the *C. albicans* genome (Gaur et al., 2008). As per the data from the *Candida Genome Database* (<http://www.candidagenome.org>), *C. albicans* genome harbors three *S. cerevisiae* homologous (with >60% similarity) *QDR* genes; *QDR1* (orf19.508), *QDR2* (orf19.6992), and *QDR3* (orf19.136). The yeast *S. cerevisiae* *QDR2* homologous MFS transporter *CgQdr2* is the only *QDR* transporter identified in *C. glabrata* conferring drug resistance, particularly to imidazoles and other stress factors, mainly by reducing the intracellular accumulation of such compounds (Costa et al., 2013).

Based on the transcript expression profiling of *C. albicans*, the *QDR1* gene is upregulated in the cells treated with the antifungal dye Malachite green (Dhamgaye et al., 2012). In *C. albicans*, *QDR* genes are essential for biofilm formation as *QDR* gene deletions can lead to defects in the architecture and thickness of biofilms (Shah et al., 2014). Our previous study suggested that the

QDR1 gene and its two paralogues, *QDR2* and *QDR3*, although do not play any direct role in drug transport but have an important role in the pathogenicity of *C. albicans* as they are involved in the regulation of various physiological responses as confirmed by their genomic, lipidomic and phenotypic profiling. (Shah et al., 2014). The present study shows that the *QDR123KO* (*QDR* triple Knockout strain) displays overexpression of glucose transporter genes (*HGT8*, *HGT9*), particularly under glucose limiting conditions. Furthermore, the overexpression of *HGT* genes results in enhanced glucose accumulation in *QDRKO*s, which is one of the known mechanisms to increase drug resistance and virulence in this human pathogenic yeast (Rodaki et al., 2009; Dikicioglu et al., 2014). Thus, the attenuation in virulence seen as the result of loss of *QDR* transporter genes can be partially overcome by increasing glucose and glycerol accumulation, leading to enhanced azole drug resistance in the *QDRKO* strains.

2. Materials and Methods

2.1 Materials, strains, and culture procedures

Most of the chemicals employed in this study were procured either from Sigma or Merck. The media components were obtained from HiMedia (Mumbai, India). The cDNA synthesis and Taq DNA polymerase kits were obtained from Thermo Scientific. The glucose (GO) assay kit (GAGO20) and glycerol estimation kit (MAK117) were purchased from Sigma Aldrich, while the RNA extraction kit was procured from Promega. The Primers employed in this study were obtained from IDT (India).

The fungal strains used in this study are listed in Supplementary Table S1. The strains were maintained as frozen stocks in cryovials and propagated either in Yeast extract peptone-dextrose (YEPD) broth or YEPD agar plates whenever required. For the growth of the strains in low, medium, and high glucose concentrations, YEPD medium containing 0.2%, 2%, and 5% D-glucose respectively, was used.

2.2 Susceptibility Assay

The susceptibility of the mutants against various drugs/stresses was analyzed by serial dilution assay as described previously (Mukhopadhyay et al., 2002; Shah et al., 2014). In this method, solid YEPD media plates containing drugs/stressors were spotted with different strains. Cells were suspended in normal saline to an optical density of 0.1 at 600 nm, followed by 5-fold serial dilutions. 4 µl samples of 5-fold serial dilutions were spotted for each strain on YEPD plates either containing drugs/stressors or only YEPD (Control plates). Finally, the plates were incubated at 30°C for 48 hrs and visualized.

2.3 RNA Isolation, cDNA Synthesis, and RT-PCR/qRT-PCR

All the strains (WT and *QDR* mutant strains) were cultured in YEPD media at 30°C up to the mid-exponential growth phase. The cells were harvested and the resulting cell pellet was washed at least three times with PBS. The final cell pellet was a starting material for RNA isolation and further experiments. RNA isolation was carried out using Promega's SV Total RNA Isolation System. RNA was eluted using nuclease-free water. This RNA obtained was quantified spectrophotometrically, separated by electrophoresis in a denaturing formaldehyde agarose gel, and visualized. According to the manufacturer's instructions, a cDNA synthesis kit was used to synthesize the complementary DNA (cDNA). 1 µg of the extracted RNA was used for cDNA synthesis. RT-PCR (reverse transcription-PCR) was performed using the standard protocol. In brief, isolated RNA (1 µg) was treated with DNase and after that primed with oligo(dT)₁₈ and random primers for cDNA synthesis at 42°C for 60 min. The reverse transcription reaction was terminated by heating the contents at 70°C for 5 min. The cDNA product (1 µl) synthesized was directly used for PCR amplification (50 µl) using gene specific forward and reverse primers as described previously (Shah et al., 2014). The *ACT1* gene was employed as an endogenous reference control. The amplified products were separated by gel electrophoresis and quantitated relative to their expression in control strains using Bio-Rad Image Lab 5.0 Software. Each reaction was performed in triplicate and mean values of relative expression were determined for

each gene. The primers and the sequence of the primers used are listed in Supplementary Table 2.

For quantitative RT-PCR (qRT-PCR), RNA was isolated by the Qiagen RNeasy kit. The isolated RNA was quantified by UV-Vis spectroscopy and the quality was checked by measuring absorbance at 260/280 nm. cDNA was synthesized using the Bio-Rad iScript cDNA Synthesis Kit. 50 ng RNA equivalent of cDNA was used per real-time reaction (25 μ l). Real-time PCR was performed in a Rotor-gene machine from Qiagen. QuantiFast SYBR green master mix (Qiagen) was used for real-time PCR. For obtaining better results, an average of triplicates were taken for analysis. $\Delta\Delta C_t$ method was used for calculating fold changes.

2.4 Quantification of intracellular metal ions/elements

The concentrations of the metals within the cell were measured by Energy Dispersive X-ray Fluorescence Technique (EDXRF). PAN Analytical Epsilon 5 EDXRF system was used in the present study. The 250 ml culture of each mutant and WT *Candida* cells were grown overnight, centrifuged, and washed with 0.1N HCl thrice to remove all the particles adsorbed to the cell surface to avoid bioadsorption of the accumulated metal can be quantified. Each cell pellet was dried in the Laboratory oven at 40°C and manually powdered using Mortar and Pestle. Each powdered sample was mixed with an equal amount of boric acid, which acted as a binding material and was applied to the presser to form pellets. Each pressed pellet was subjected to elemental analysis and individually analyzed by EDXRF.

2.5 Glucose Quantification

Briefly, secondary culture ($OD_{600}=0.1$) was set up in 50ml YEPD containing 0.2% and 2% glucose for all the tested strains (*QDRKO*s and WT strains). The cells were harvested and the resulting cell pellet was washed with sterilized double-distilled water. The cell pellet was used to determine the total glucose content inside the cells. The cell pellet was dried (at 30°C for a few hours), weighed (in mg), and disrupted using glass bead-assisted cell disruption. The samples were then extracted with deionized water. The glucose content in the samples was determined

using Glucose (GO) assay kit using the standard glucose control and following the manufacturer's recommended protocol. To determine the amount of glucose (in mg); the following equation was used:

$$\text{mg Glucose} = (\Delta A_{540} \text{ of Test}) (\text{mg Glucose in Standard}) / \Delta A_{540} \text{ of Standard}$$

The mg glucose determined above was multiplied by the dilution factor made during sample preparation to get the final glucose concentrations accumulated per mg dry weight of different samples.

2.6 Glycerol Accumulation Assay

Glycerol concentration in the samples was determined using a Glycerol assay kit (MAK117, Sigma-Aldrich), following the manufacturer's recommended protocol. Briefly, secondary culture ($OD_{600}=0.5$) was set up in 5ml YEPD containing 0.2% and 2% glucose for all the tested strains (*QDRKO*s and WT strains). After 2 hours of incubation, 1M NaCl (final concentration) was added to the culture. Following NaCl treatment, 1 ml cell suspension of each culture strain was taken in two separate microfuge tubes pre-labeled as Extracellular and Total. The microfuge tubes labeled as Extracellular were centrifuged, and the supernatants were boiled at 100°C for 10 minutes in a dry bath incubator and then cooled on ice for about 5-10 minutes. In the case of the tubes labeled as total, the cell suspensions were first boiled at 100°C for 10 minutes, cooled, and then centrifuged to obtain the supernatant. 10 µl of each sample (supernatant) was taken, and 100 µl of the reaction mixture (containing 96 µl assay buffer, 1 µl ATP, 1 µl dye, and 2 µl enzyme) was added to it in 96 well-plate. After incubation for 10 to 15 minutes, the absorbance was measured at 570 nm using a microplate reader. Finally, the standard curve was prepared, and the glycerol content was calculated. The extracellular glycerol values were subtracted from the total glycerol values to obtain the values for the intracellular glycerol content.

2.7 Proteinase Activity Assay

For proteinase activity assay, YBD medium containing yeast extract, Bovine Serum Albumin (BSA), and glucose were prepared in sterilized Petri dishes. Briefly, all the tested strains (*QDRKO*s and WT strains) were cultured overnight in YEPD medium at 30°C. 5 µl (OD₆₀₀=0.1) of the overnight cell culture of each sample was inoculated onto the YBD plates. The plates were then incubated for four days at 37°C. The proteinase activity was determined by measuring: 1) the diameters of the colonies and 2) the diameters of the clearing zone around the colonies. The proteinase activity of each sample was represented by calculating the ratio of the diameter of the precipitation zone plus colony diameter to the diameter of the colony only.

2.8 Morphogenetic study at different glucose concentrations

The strains (*QDRKO*s and WT strains) were cultured overnight in 5 ml YEPD media at 30°C up to the mid-exponential growth phase. The cells were harvested and the resulting cell pellet was washed at least three times with PBS. The washed cells were mixed with sterilized double-distilled water (500 µl) and starved at 30°C for 2 hours. After starvation, OD₆₀₀ was adjusted to 1.0 and the required amount of cells (250 µl; OD₆₀₀=1.0) were then inoculated into fresh, sterilized microfuge tubes containing 1ml YEPD medium with different glucose concentrations (0.2%, 2%, and 5% D-glucose) supplemented with 2% Fetal Bovine Serum and incubated at 37 °C for 2 hours. Finally, cells were visualized for hyphae formation with the aid of microscopy.

2.9 Statistical analysis

To determine the statistical significance of the differences in the data, Student's *t*-tests were performed where $P < 0.05$ was considered statistically significant.

3 Results

3.1 Transcript profiling analysis reveals the differential expression of genes in *QDRKO* strains belonging to different pathways:

Considering the importance of *QDR* genes in the virulence and/or pathogenicity of *C. albicans* as reported earlier (Nunes et al., 2001; Vargas et al., 2004; Tenreiro et al., 2005; Shah et al., 2014), we focused on other pathways where *QDR* genes might be involved directly or indirectly and thus leading to altered gene expressions or phenotypic differences. The analysis of transcripts data (GEO dataset accession number GSE41535) (Shah et al., 2014) revealed that many genes belonging to different categories show differential expression patterns in *QDRKO* strains. Some of the categories of genes exhibiting differential expression patterns and taken up for further study include chitin and β -glucan synthases, mannosyl transferases, vacuolar genes, ion transporters, acid phosphatase genes, and different high-affinity glucose transporters (**Table 1 A-F and Figure 1 A-C**). Semi-quantitative RT-PCR of a few randomly selected differentially expressed genes (HGT8 and 9, CCC2, CHS6, VPS33, and MNN7) was performed to validate the transcriptomics data further. The results obtained from semi-quantitative RT-PCR were mainly consistent with the differential expression of genes observed in our data (**Supplementary Figure 1**).

3.2 Cell wall (CW) stresses and altered ion transporter gene expression in *QDRKO* strains did not lead to significant phenotypic defects:

Since we were able to identify a few selected categories of genes like cell wall and ion transporter genes etc. showing differential expression patterns due to *QDR* gene deletion, we analyzed the *QDR* deleted mutants exposed to cell wall perturbing agents (calcofluor-white, caspofungin), high salt and heavy metal stress (NaCl, MnCl₂, CdCl₂, LiCl, etc.), different pH (pH 8, pH 3.5), different temperature (25° C, 42° C), etc. As can be comprehended from the susceptibility spot assays (**Figure 2 A & B**), none of the mutant cells displayed any significant change in the susceptibility patterns compared to the WT strain on exposure to any of the tested stresses. Considering the differential expression profiles of different cell surface proteins and membrane localization of these *QDR* transporters, we examined the cell surface morphology of these mutants by scanning electron microscopy. Scanning electron microscopy pictures also did not show any significant change in the surface morphology (**Supplementary Figure 2**).

Since many ion transporter genes also show significant alteration in their expression levels in *QDRKO* strains, we also investigated these knockouts for their intracellular ion/elemental (e.g., calcium, sulfur, zinc, copper, etc.) accumulation levels by Energy Dispersive Xray Fluorescence (EDXRF) Spectrometer. The results shown in the **Supplementary Figure 3 (a-d)** demonstrate that the intracellular contents of these elements in *QDR* deletion mutants do not show any significant changes as compared to the WT strain despite the differential expression pattern shown by some of the ion/ metal transporter genes in case of the *QDRKO* strains.

Together, these results suggest that in the *QDRKO* strains, the altered expression of genes encoding cell surface proteins, vacuolar proteins, ion transporters, acid phosphatases, and other such genes showing differential expression in transcripts profiling data did not lead to any significant and visible phenotypic changes.

3.3 *QDRKO* strains show decreased proteinase activity

While the WT and all the *QDRKO Candida* strains exhibited proteinase activity, the proteinase activity significantly differed. The knockout strains displayed lower proteinase activity compared to the WT strain. The average proteinase activity of WT *Candida* strains was equal to (3.3 cm) while the proteinase activity of *KO* strains ranged from (1.4 to 1.63 cm) (**Table 2 and Figure 3**). All the *QDRKO* strains showed decreased proteinase activity compared to the WT *C. albicans* strain. The difference in the proteinase activity points towards the significance of *QDR* genes in the modulation of proteinase activity which can be one of the reasons for decreased virulence in these otherwise pathogenic *Candida* species.

3.4 Transcript levels of *HGT8* and *HGT9* genes in response to different glucose levels being affected by *QDR* genes

The high-affinity glucose transport genes *HGT8* and *HGT9* show upregulation (more than two-fold) in *QDR123KO* compared to the WT strain in the transcript profiling data. We selected *QDR123KO* and WT strains to check and compare the expression levels of these *HGT* genes at different glucose concentrations to find out if the glucose concentrations present in the growth

medium could impact the *HGT* gene expression in the presence or absence of *QDR* genes. We used three different glucose concentrations, low (0.2%), medium (2%) and high (5%). To obtain the transcription profiles, the strains were grown at these three glucose concentrations. The transcription profiles of these *HGT* genes revealed that both *HGT8* and *HGT9* genes were expressed at low (0.2%) glucose concentration in *QDR123KO* and the WT strain *SC5314*, but the expression of both *HGT8* and *HGT9* was marginally higher in the case of *QDR123KO* strain as compared to the WT strain (**Figure 4 A & B**). The enhanced expression levels of *HGT8* and *HGT9* in the *QDR123KO* strain at low glucose concentration suggest that the loss of *QDR* genes leads to increased expression of glucose transporter genes, particularly at a low glucose concentration of extracellular media.

As far as the transcription profiles of *HGT8* and *HGT9* genes at other glucose concentrations is concerned, the results revealed that the *HGT8* gene expression level was high in *QDR123KO* compared to its expression levels in WT strain at all the three tested glucose concentrations (**Figure 4 A & B**). While the *HGT9* gene expression levels in the *QDR123KO* strain were elevated significantly at lower glucose concentrations only compared to its expression levels at 5% glucose concentration at which the expression levels of *HGT9* were not significantly different between WT and *KO* strains (**Figure 4B**). Thus, it seems that the expression of *HGT8* gets elevated in *QDR123KO* at all tested glucose levels, while the expression of *HGT9* shows elevation only at low glucose levels in the media. This differential expression pattern points to the importance of *HGT8* gene overexpression in the *QDR123KO* background at all the tested media glucose concentrations. In contrast, *HGT9* gene overexpression is maintained only when low glucose levels are in the media. Furthermore, the quantitative RT-PCR (qRT-PCR) analysis also revealed similar expression patterns of *HGT8* and *HGT9* genes in the *QDR123KO* and WT strains at low media glucose levels (**Figure 4C**). qRT-PCR expression analysis was done only at lower glucose levels since the above transcriptional profile shows considerable differences only at lower glucose levels.

3.5 *QDRKO* strains show increased glucose accumulation:

In most heterotrophic organisms, glucose represents the primary carbon and energy source utilized to regulate several processes, including growth, development, and metabolism (Özcan et al., 1999; Dos Reis et al., 2013). Many studies have elucidated the role of glucose in regulating growth and virulence processes in different yeast species, including the pathogenic fungi *C. albicans* (Brown et al., 2006; Rodaki et al., 2009; Dos Reis et al., 2013). Our current study has also found the differential expression pattern of various sugar transporter genes in *QDRKO* *Candida* strains, as seen in transcriptomic data and subsequent validation above.

Considering the role and the effect of glucose on the growth, development, and survival processes on organisms like *C. albicans* and the differential regulation of glucose transporter genes in *QDRKO* strains, we explored the possible involvement of *QDR* transporters and glucose in the survival strategy adopted by *C. albicans*. We investigated the *QDRKO* strains for the intracellular glucose accumulation patterns to test this hypothesis. The glucose accumulation assay of the *QDRKO* strains revealed a significantly higher accumulation of glucose in all four knockouts, with the highest accumulation recorded in the *QDR123KO* strain compared to the WT strain SC5314. Furthermore, glucose accumulation was determined only at two lower media glucose levels (0.2% and 2%) since significant modulation in *HGT* gene expression was seen at lower media glucose levels only. As shown in **Figures 5A and B** enhanced glucose accumulation was seen both at low and average glucose concentrations of the media. However, glucose accumulation levels were higher at 2% media glucose than accumulation levels seen at 0.2% media glucose even though the *HGT* gene expression profile was more or less similar, possibly due to the differences in the availability of the extracellular media glucose.

Based on these results, we concluded that the *QDR123KO* strain showed elevated expression levels of *HGT* genes (*HGT8* & *HGT9*) and showed enhanced glucose accumulation levels. Thus, the loss of *QDR*/*MFS* transporter genes likely leads to the upregulation of specific glucose *MFS* transporter genes, which concurrently leads to increased glucose accumulation.

3.6 *QDRKO* *Candida* strains show increased resistance to azole antifungals at lower glucose concentrations compared to WT *Candida* strain

We further evaluated and compared the susceptibility of the *QDRKO* and WT strains grown at different glucose concentrations (low, medium, and high) to different classes of antifungal drugs, including amphotericin B (a polyene), caspofungin (an echinocandin), ketoconazole (an imidazole), fluconazole and voriconazole (triazoles). From drug susceptibility spot assays and MIC determinations, we were able to find some interesting responses shown by *QDRKO* strains against different antifungals at different glucose concentrations. All the knockout strains compared to WT strain grown at lower glucose concentrations were comparatively resistant to all the tested azoles, Ketoconazole, Voriconazole, and Fluconazole (**Fig 6 A-C**). Susceptibility to the drugs amphotericin B (a polyene) and caspofungin (an echinocandin), on the other hand, showed no difference between the knockouts and WT strain at all three tested concentrations of glucose (**Fig 6 D**). The results obtained from drug susceptibility spot assays were also corroborated by MIC determinations (**Table 3**). Our results conclude that different concentrations of the carbon source (glucose) significantly and specifically impact the drug resistance attributes of *C. albicans* as displayed by the tested *QDRKO*s and WT strains (**Fig 6 A-C**). Surprisingly this resistance pattern shown by *QDR123KO* was not additive as the resistance pattern shown by individual and multiple gene deletions were more or less similar.

Additionally, the growth pattern displayed by *QDRKO* strains also varied with different azole drug concentrations and was not uniform. In the case of azoles, overall growth was better at intermediate drug concentrations used (**Fig 6**). Overall, these results suggest that *QDRKO*s at lower glucose concentrations are mainly resistant to azoles than the WT strain possibly, due to enhanced glucose accumulation seen above in knockout strains. This again reflects that *QDRKO* strains impart enhanced drug resistance via increased glucose accumulation which could partially compensate for the cellular resistance/virulence loss seen in the absence of the *QDR* MFS transporter proteins.

3.7 Glucose concentrations modulate the morphological switching phenomenon and glycerol accumulation in *QDRKO* strains

QDRKO strains were evaluated for hyphae formation at different glucose concentrations of hyphae-inducing media. As shown in **Figure 7**, *QDRKO* strains showed significant differences in the hyphae formation compared to the WT *Candida* strain at different glucose concentrations. Compared to the WT strain, all the knockout strains show increased hyphae formation, which was more pronounced at lower glucose concentrations than that seen at higher glucose concentrations. Modulation in hyphal morphogenesis between *QDRKO* and WT strains again signifies the vital role of *QDRs* in maintaining cellular pathogenicity/virulence. In the absence of the *QDR* MFS transporters, where virulence is decreased, enhancing the hyphal morphological forms seems to help the *Candida* strains maintain considerable pathogenicity.

Accumulation of metabolites like glycerol inside the yeast cells is considered as one of the mechanisms to evade the stress response (Dikicioglu et al., 2014). Considering the important role of such metabolites in maintaining stress resistance, we also checked the glycerol accumulation in *QDRKO*, and WT *Candida* strains at medium (2%) and low (0.2%) glucose concentrations of extracellular media. The glycerol accumulation assay of the *QDRKO* strains revealed a significantly higher glycerol accumulation for all the four *QDRKO* strains (*QDR1KO*, *QDR2KO*, *QDR3KO*, and *QDR123KO*), both at medium and lower media glucose levels. The highest accumulation was observed for the *QDR* triple KO (*QDR123KO*) strain compared to the WT strain SC5314 (**Figure 8 A & B**). Interestingly, the phenomenon was seen to be more pronounced at low media glucose concentration (**Figure 8B**). The results demonstrating higher intracellular glycerol accumulation in *QDRKO* strains highlight one more possible way adopted by *Candida* cells to neutralize the defects incurred due to the absence of *QDR* genes.

4. Discussion

Currently, fungal infections represent one of the major challenging diseases in humans (Kohler et al., 2015). These infections remain serious and underappreciated causes of illness and death. Fungal infections historically have been less-recognized and challenging to detect, and treatment options are poor. The reasons for their emergence are likely multifactorial (Brandt and Park,

2013). Moreover, the invasive fungal infections often accompanied by the MDR phenomenon caused by pathogenic fungi are often considered a burden on socioeconomic growth, leading to high disease and death rates. Drug transporters (ABC/MFS) have been known to play an essential role in the process of drug resistance in human pathogenic fungi. A lot of work has been done on ABC transporters compared to MFS transporters. MFS transporters belonging to the *QDR* subfamily have been the main focus of this study. Considering the essential role of *QDR* genes in different pathogenic attributes of human pathogenic fungi *C. albicans* (Shah et al., 2014), the current study underlines the role of *QDR* genes in survival and virulence/pathogenicity of *C. albicans*.

Given that *QDR* transporters impact biofilm development and virulence, we hypothesized that *QDR* genes might directly or indirectly affect different pathogenicity and virulence traits via regulating different associated pathways and processes. Keeping this in consideration, we assessed the role of other cellular pathways (which show differential expression patterns upon the deletion of *QDR* genes), which can assist the cells in overcoming stress and maintaining cellular pathogenicity in the absence of *QDR* MFS transporters. *QDR* mutants show considerable differences in the expression patterns of various genes belonging to different categories (**Table 1**). Following the differential expression pattern of the various categories of genes, we analyzed the mutant phenotypes under different types of cellular stresses. We found that the *QDRKO* and WT strains showed comparable susceptibility patterns to various stresses from phenotypic assays. The cell surface morphology of these mutants also did not reveal any significant change, as all these knockout and WT *Candida* cells showed similar surface morphology. Energy Dispersive X-ray Fluorescence (EDXRF) Spectrometry results further explained that the intracellular ion/elemental accumulation levels in *QDR* mutants do not display any significant alterations as compared to the WT strain despite the differential expression patterns shown by many of the ion/ metal transporter and related genes in the *QDRKO* strains.

Notwithstanding the phenotypes observed above, *QDRKO* strains show considerable differences in glucose and glycerol accumulation. *QDR*-deletion strains of *C. albicans* show significantly higher glucose accumulation among all the tested *QDRKO* strains. To our interest, the highest

accumulation was displayed by the *QDR123KO* strain compared to the WT strain. Our results conclude that the *QDRKO* strains show higher glucose accumulation, likely to compensate for the loss of QDR/MFS transporter genes; by the upregulation of some specific glucose MFS-transporter genes. Thus, *QDR* and glucose MFS transporters may share a contributing and interdependent role in assisting *C. albicans* in acquiring the necessary survival and optimal virulence strategy (**Figure 9**). In the case of yeast *S. cerevisiae*, it has also been reported earlier that *QDR3/PDR3* mutants display overexpression of a low-affinity hexose transporter (*HXT3*) under glucose abundance and overexpression of some high-affinity glucose transporters (*HXT2*, *HXT4*, and *HXT7*) under glucose limitation (Dikicioglu et al., 2014).

Consequently, as a pre-adaptive and protective mechanism to overcome the loss of drug resistance-related genes *PDR3/QDR3* in yeast cells, there is an intracellular accumulation of glucose and glycerol and inorganic phosphate as an ion pool to stabilize the two accumulated metabolites (Dikicioglu et al., 2014). As far as the expression of the high-affinity glucose transporter genes in *QDRKO* strains is concerned, the transcripts profiling results revealed the differential expression patterns of these *HGT* genes in *QDRKO* strains. Both *HGT8* and *HGT9* genes showed upregulation (more than two-fold) in the *QDR123KO* strain compared to the WT strain and their expression varies in response to different glucose levels present in the medium. Three different glucose concentrations, low (0.2%), medium (2%), and high (5%), were used for the study and the strains (*QDR123KO* and WT strains) were grown at these glucose concentrations. Interestingly, in individual *QDRKO*s, the expression pattern of these *HGT*s as obtained from transcriptomics data was not the same as seen in *QDR123KO*. The possible reason can be the compensation of the functional loss of individual *QDRKO* by other *QDR*s present in the cell, which was not possible in the triple knockout strain.

In many studies, the expression of several high-affinity glucose transporter genes in WT *Candida* strains has been known to increase at low sugar levels (0.2%) (Van Ende et al., 2019). Interestingly, our study has also witnessed a similar trend, but the expression levels were further elevated in the *QDR123KO* strain compared to WT cells. Our *HGT* transcription profiles revealed that both *HGT8* and *HGT9* genes were expressed at low (0.2%) glucose concentration

both in *QDR123KO* strain and WT strain and the expression of both genes was higher in the case of *QDR123KO*. These higher expression levels in the *QDR123KO* strain at low glucose concentration suggest that the loss of *QDR* genes leads to further increased expression of glucose transporter genes. Some previous studies have reported the expression of different *HGTs* in *C. albicans* in response to different glucose levels in the medium (Fan et al., 2002), while our current study explains the *HGT* gene expression patterns in *C. albicans* concerning *QDR* genes, thereby pointing to the possible role of glucose-based/sensed pathways and processes in aiding the survival and growth of *C. albicans*. The results suggest the interplay of quinidine drug resistance transporters and glucose transporters as one of the possible mechanisms employed by *C. albicans* in maintaining drug resistance and pathogenicity (**Figure 9**). Expression profile results of *HGT8* and *HGT9* genes at different glucose concentrations revealed high *HGT8* gene expression in *QDR123KO* compared to its expression levels in WT strain at all three tested glucose concentrations. *HGT9* expression levels in the *QDR123KO* strain, on the other hand, displayed significant elevation at lower glucose concentrations only. Herein, we conclude that *HGT8* expression is elevated in *QDR123KO Candida* strains compared to its expression in WT *Candida* strains, irrespective of the glucose levels present in the media.

In contrast, *HGT9* expression displayed elevation at lower media glucose levels only. Multiple studies have revealed the importance of glucose for survival and pathogenicity in different yeast species, including the pathogenic fungi *C. albicans* (Brown et al., 2006; Rodaki et al., 2009; Dos Reis et al., 2013). Thus, the overexpression of glucose transporters and subsequent glucose accumulation seems to be a probable effort made by *Candida* cells to partly compensate for the loss of *QDR* genes, which is even more pronounced at the low glucose concentration of extracellular media. This can be one of the important contributing factors maintaining drug resistance up to certain levels of drug stress.

Additionally, in yeast cells, including the pathogenic yeast *C. albicans*, intracellular glycerol accumulation has been found crucial for the maintenance of osmolarity, and consequently, the defects in glycerol homeostasis can have a serious impact on the morphogenetic plasticity and pathogenicity of the organism (Fan et al., 2005; Dikicioglu et al., 2014). This study also

examined the accumulation of glycerol in *QDRKO* and WT *Candida* strains at different glucose concentrations of the extracellular media. Interestingly, the *QDRKO* strains displayed significantly higher glycerol accumulation than the WT strain. This higher intracellular glycerol accumulation in the absence of *QDR* genes could be one of the possible reasons for exhibiting strong osmotolerance. Thus, in the absence of *QDR* transporters, the activity of glucose transporters and subsequent accumulation of glucose and glycerol helps in improving the adaptation of the organism to different stress responses in *C. albicans*.

It has been reported that the activity of different antifungal agents like Voriconazole and Amphotericin B in *C. albicans* and *C. tropicalis* is compromised on increasing the glucose concentration in the culture medium (Mandal et al., 2014). In our study, *QDRKO* strains at lower glucose concentrations were mainly resistant to azole antifungals than the WT strain, consistent with the upregulation of different *HGTs* and subsequent increase in intracellular glucose levels. Moreover, the *QDRKOs* at low glucose concentration also showed increased hyphae formation, augmenting the enhanced resistance attributes observed in *QDRKOs* at low glucose concentration. The study guides our understanding of the glucose-based/sensed mechanisms and pathways in *C. albicans* that could be unraveled further as a potential drug target.

Glucose exposure has been shown to provide cationic and oxidative stress resistance and also enhance drug resistance in *C. albicans* cells (Rodaki et al., 2009) which again points to the fact that we could not see the altered susceptibility to such stresses as there is increased accumulation of glucose in the *QDRKO* strains. In contrast to non-diabetic patients, in diabetics, the frequency of occurrence/colonization of *Candida* spp. is much higher, with *C. albicans* being the most frequently available species and any change in the physiological factors like glucose and pH positively impacts the *Candida* flora growth during oral candidiasis in people with diabetes (Mohammadi et al., 2016). The growth of *C. albicans* cells is directly affected by the glucose concentration in diabetic patients. With such poor glucose management, the patients are more prone to *C. albicans* infections (Man et al., 2017). Further, our knockouts displayed higher resistance to different azole antifungals at low glucose levels of media, pointing towards the modulation of basic metabolism in the knockouts concerning glucose mismanagement. In human

cancer cells, extracellular glucose levels can activate several survival pathways under glucose-deprived conditions and enhance glycolysis via increasing GLUT1 levels to gain drug resistance (Bhattacharya et al., 2014). From the above discussion point of view, it becomes clear that glucose plays a crucial role in the survival strategy of *C. albicans* and thus could have a significant role in the virulence and pathogenicity of *C. albicans* when exposed to different microenvironments inside human cells. Although the specific communication mechanism between two groups of MFS transporters may not be precise yet, based on our results, we hypothesize that the deletion of *CaQDR* genes leads to an increase in the expression of glucose transporter genes. Although the difference in expression profiles of *HGTs* in *QDRKO*s is subtle but looks significant enough to possibly modulate the intracellular glucose accumulation, which regulates (directly/indirectly) the accumulation of other important metabolites like glycerol. The differences in transporter expression, glucose accumulation, glycerol accumulation, and other related phenotypes collectively seem to partially compensate for the altered pathogenicity and virulence caused by the loss of *QDR* genes. These *QDRKO* strains, compared to the WT strain, especially at lower glucose concentrations, show an interesting drug resistance pattern in the absence of the *QDR* MFS transporters. The accumulation of metabolites as a stress response represents an important parameter to be analyzed to evaluate the process of drug resistance and virulence in pathogenic *Candida* species. Thus intracellular metabolite levels can be important therapeutic targets and approaches for designing novel metabolome-based antifungals. The resistance pattern shown by *QDRKO* and WT strains against different azole antifungals emphasizes the selective use of drugs and drug concentrations to find the most active and effective drug concentrations. Such selective drug concentrations can be more effective and leave fewer chances for developing antifungal drug resistance, which has become a menace under the current antifungal drug therapy regime. Moreover, a better understanding of the regulation of the *QDR* genes is required, as their deletion seems to be one of the mechanisms in developing resistance in *C. albicans*.

Our findings based on expression profiling of *C. albicans QDR KO* stains and subsequent studies could serve as investigative data and a basis for future studies in elucidating the complex interplay mechanism between quinidine drug resistance transporters and glucose transporters in

maintaining drug resistance and pathogenicity in this human pathogenic yeast. Furthermore, considering glucose as one of the central molecules in metabolism, advanced research investigating the interrelationship of the *QDR* and *HGT* genes will further help in elucidating the effective role of these genes in the physiological, molecular, and biochemical processes of *C. albicans* and thereby will enhance our understanding of the absolute role of *QDR* genes and *HGT's* in *Candida* pathogenicity.

5. Conclusions: In this study, we were able to show that *QDR* knockout *Candida* strains show a significantly higher accumulation of glucose and glycerol than the WT strain. We conclude that the *QDRKO Candida* strains upregulate the expression of some high-affinity glucose MFS-transporter genes show higher metabolite accumulation and modulation in a few other pathogenicity factors as a stress response, likely to compensate for the loss of QDR-MFS transporter genes. This metabolite accumulation culminates in enhanced azole drug resistance reflected by *QDRKO* strains, particularly at low extracellular glucose concentrations. Thus, *QDR* and glucose MFS transporters share a contributing and interdependent role in assisting *C. albicans* in acquiring the necessary survival and optimal pathogenicity.

Author contributions

Hafsa Qadri: Investigation; Data curation; Methodology; Writing - original draft, **Abdul Haseeb Shah:** Investigation; Conceptualization; Funding acquisition; Supervision; Writing - original draft, **Munazah Fazal Qureshi:** Methodology; Validation; Formal analysis, **Manzoor Ahmad Mir:** Visualization; Writing - Review & Editing, **Rajendra Prasad:** Conceptualization; Writing - Review & Editing.

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Declaration of competing interests

The authors declare no conflict of interest

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Supplementary Data

Table S1 and S2, Figs. S1-S3

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Figure Captions

Figure 1: Genes belonging to different categories showing differential expression profile (more than two-fold change) in QDRKO strains A) Mannosyl Transferases or related genes, B) Chitin and β -Glucan Synthases or related genes C) Vacuolar, Ion, Sugar Transporter Genes and acid phosphatases.

Figure 2: The susceptibility spot assay results of QDR KO strains in the presence of (A) Metal/ion stress/chelation (B) Cell wall and temperature stress, Acid/oxidative stress. Fivefold serial dilution of 0.1 OD of each strain was spotted on control/drug plates.

Figure 3: Estimation of proteinase activity. Proteinase activity of *QDRKO*s (*QDR1KO*, *QDR2KO*, *QDR3KO*, and *QDR123KO*) VS WT strains of *C. albicans*, on the solid YBD medium containing yeast extract, BSA, and glucose. The proteinase activity of the strains could be seen as the development of respective transparent zones around each strain.

Figure 4: Impact of extracellular glucose levels on the expression of (A) *HGT8* gene and (B) *HGT9* gene in *QDRKO* strains compared to WT strain. For obtaining the transcription profiles, the strains were grown at three different glucose concentrations low (0.2%), medium (2%), and high (5%). C) Validation *HGT* gene overexpression in *QDR123KO* strain by qRT-PCR (both at 0.2% and 2% glucose) compared to expression levels in WT strain *SC5314*. The y axis shows the fold change (FC) in expression levels of respective *HGT* genes in *QDR123KO* relative to expression levels in WT strain (taken as 1). The results are the mean ($\pm$ S.D) of at least three independent experiments.

Figure 5: Glucose accumulation levels in QDR knockout strains compared to the WT strain of *C. albicans*. A) Glucose accumulation at 2% glucose levels of media and B) Glucose accumulation at 0.2% glucose levels of media. The glucose content in the samples was determined by comparing it to the standard glucose control provided in the Glucose (GO) assay kit. The results are the mean ($\pm$ S.D) of at least three independent values.

Figure 6 (A- D): Susceptibility spot assay of the QDRKO and WT strains of *C. albicans*. Different strains were spotted on YEPD medium containing different glucose concentrations (0.2% D-glucose, 2.0 % D-glucose and 5.0 % D-glucose) against different classes of antifungals viz: **A) Voriconazole** (VORI; 0.025µg/ml, 0.05µg/ml, 0.1µg/ml, and 0.2µg/ml), **B) Ketoconazole** (KTC; 0.05µg/ml, 0.1µg/ml, 0.15µg/ml and 0.2µg/ml), **C) Fluconazole** (FLC; 1.5µg/ml, 2µg/ml and 2.5µg/ml), **D) Amphotericin-B** (AMP; 0.25µg/ml and 0.5µg/ml) and **Caspofungin** (CSF; 0.25µg/ml and 0.3µg/ml). All the images are representative of 3 to 4 individual experiments obtained after 48 hours of incubation at 30 °C.

Figure 7: Morphological differences in hyphae formation in QDRKO and WT strains of *C. albicans*. Briefly, required cells ($OD_{600}=1.0$) were inoculated into a YEPD medium containing different glucose concentrations (0.2% D-glucose, 2.0 % D-glucose, and 5.0 % D-glucose) supplemented with 2% Fetal Bovine Serum and incubated at 37 °C for 2 hours. Finally, the cells were visualized for hyphae formation with the aid of microscopy

Figure 8: Glycerol accumulation levels in QDR knockout strains compared to the WT strain of *C. albicans*. A) At 2% media glucose content and B) At 0.2% media glucose content. The glycerol content in the samples was determined by comparing it to the standard glycerol control provided in the Glycerol estimation kit. The results are the mean ($\pm$ S.D) of at least three independent values.

Figure 9: Schematic illustration of the impact of the deletion of QDR MFS transporter genes on the survival of *C. albicans*. Deleting *QDR* genes (individually/collectively) led to the modulation of *HGT* gene (glucose MFS transporters) expression resulting in an enhanced glucose accumulation. The crosstalk between these two classes of MFS-transporters is also reflected in enhanced hyphal morphogenesis, glycerol accumulation, etc., collectively leading to increased azole drug resistance and possibly for maintaining virulence/pathogenicity.

Supplementary Figure 1: Validation of the transcriptomics data using semi-quantitative RT-PCR. Few genes selected randomly for the study include HGT8 and 9, CCC2, CHS6, VPS33, and MNN7.

Supplementary Figure 2: Representation of the Scanning Electron Microscopy (SEM) images of the WT strain SC5314 and QDR deletion mutants displaying their respective growth and surface morphology.

Supplementary Figure 3: Intracellular ion contents in QDR deletion mutants and WT strains of *C. albicans* represented either ppm or percentage of total ion abundance. The results are the mean ($\pm$ S.D) of at least three independent values.

Table 1: Differential expression patterns (fold change greater than 2 ($\text{Log}_2 >1$ or <-1)) of different categories of genes a) Chitin, Beta-glucan synthase and related genes b) Mannosyl transferase and related genes c) Vacuolar genes d) Ion transporter and related genes e) Acid phosphatase genes and f) Sugar transporter genes. Up and down arrows indicate up and downregulation of genes, respectively.

A)

Genes Upregulated/Downregulated	QDR1 KO	QDR2 KO	QDR3 KO	TRIPLE KO
<i>CHT1</i>	1.832			
<i>KRE5</i>	1.693		1.99	
<i>CDA2</i>				1.333
<i>CHS1</i>		-1.987	-1.61	
<i>CHS6</i>		-2.130	-2.32	
<i>CHS3</i>	-1.206		-1.48	-2.392
<i>CHS8</i>			-1.55	
<i>GSL2</i>			-1.59	
<i>PIR32</i>				-2.717

745 **B-i)**

Genes Upregulated	QDR1 KO	QDR2 KO	QDR3 KO	TRIPLE KO
<i>MNT3</i>	1.876			
<i>MNN11</i>	1.800			
<i>ALG9</i>		2.05		
<i>ALG7</i>		1.64		
<i>KTR4</i>			1.69	
<i>PMI1</i>			1.58	
<i>PMT5</i>			1.45	
<i>VRG4</i>			1.40	
<i>MNN3</i>			1.34	
<i>MNN9</i>			1.33	
<i>MNN7</i>				4.138
<i>ORF19.3996</i>				1.119

746 **B-ii)**

Genes Downregulated	QDR1 KO	QDR2 KO	QDR3 KO	TRIPLE KO
<i>BMT3</i>	-1.630	-1.465		
<i>MNN1</i>	-1.929	-1.699		
<i>MNT4</i>	-1.405	-1.572	-2.07	
<i>MNN4-4</i>		-2.380	-1.26	-1.356
<i>ANP1</i>		-2.136		-1.614
<i>GAL-102</i>				-1.596
<i>HOC1</i>				-1.163

<i>IPT1</i>			-1.85	
<i>ALG11</i>			-1.89	
<i>ECM-39</i>		-1.744		
<i>ORF.3688</i>		-1.154		

747 C)

Genes Upregulated /Downregulate	<i>QDR1 KO</i>	<i>QDR2 KO</i>	<i>QDR3 KO</i>	<i>TRIPLE KO</i>
<i>VPS34</i>		1.75	1.53	
<i>PHM5</i>		1.26	1.28	
<i>VPS33</i>		-2.120	-1.77	
<i>VPS8</i>		-1.232		-1.479
<i>VPS23</i>		-1.221		-1.411
<i>CUP5</i>			1.29	
<i>VPS41</i>			-1.25	
<i>VPS51</i>	1.605			

748

749 D)

Genes Upregulated Downregulated	<i>QDR1 KO</i>	<i>QDR2 KO</i>	<i>QDR3 KO</i>	<i>TRIPLE KO</i>
<i>CCC1</i>		-1.945		
<i>CCC2</i>	2.021	-1.468	-2.28	
<i>MAC1</i>		1.49		3.204
<i>ARR3</i>	5.930		5.49	

<i>ZRT1</i>			3.61	
<i>CTR1</i>			2.09	

750 E)

Genes Upregulated/ Downregulated	<i>QDR1 KO</i>	<i>QDR2 KO</i>	<i>QDR3 KO</i>	<i>TRIPLE KO</i>
PHO89		-2.673		
PHO113		-1.267	-1.77	-2.738
PHO81				
PHO84			1.769	-1.168
PHO114	-1.076			

751 F)

Genes Upregulated/ Downregulate	<i>QDR1 KO</i>	<i>QDR2 KO</i>	<i>QDR3 KO</i>	<i>TRIPLE KO</i>
<i>HGT9</i>	-3.243	-2.111	-2.07	2.979
<i>HGT8</i>				2.831
<i>HGT18</i>		-2.012		
<i>HGT12</i>			-2.30	

752 Table 2: Proteinase activity values of the *QDRKO* vs. WT *Candida* strains

S.No	Strain	Colony diameter (cm)	Total diameter (cm)	Enzyme activity*	Relative enzyme activity (to WT)
1.	<i>QDR1KO</i>	1	1.5	1.5	0.45
2.	<i>QDR2KO</i>	1.1	1.8	1.63	0.49
3.	<i>QDR3KO</i>	1	1.4	1.4	0.42

4.	<i>QDR123KO</i>	1.2	1.8	1.5	0.45
5.	WT	0.9	3	3.3	1

753 *Total diameter equals the diameter of the precipitation zone plus the diameter of the colony.*

754 **Total diameter upon colony diameter.*

755 **Table 3: MIC₈₀ values of various drugs against *QDR KO Candida* strains A) at 0.2%**
756 **glucose concentration B) at (2%) glucose concentration.**

757 **A)**

0.2% Glucose	VORI	KTC	FLC	AMP	CSF
<i>QDR1KO</i>	1	2	10	0.125	0.0078
<i>QDR2KO</i>	1	2	10	0.125	0.0078
<i>QDR3KO</i>	1	2	10	0.125	0.0078
<i>QDR123KO</i>	1	2	10	0.125	0.0078
WT	0.03125	0.015625	1.25	0.125	0.0156

758 **B)**

2% Glucose	VORI	KTC	FLC	AMP	CSF
<i>QDR1KO</i>	0.015625	0.0078	1.25	0.25	0.125
<i>QDR2KO</i>	0.015625	0.0078	1.25	0.25	0.125
<i>QDR3KO</i>	0.015625	0.0078	1.25	0.25	0.125
<i>QDR123KO</i>	0.015625	0.0078	1.25	0.25	0.125
WT	0.0039	0.0019	1.25	0.25	0.25

759 **Supplementary Table S1: List of strains used in the study:**

Strain	Description	Reference
<i>SC5314</i> (WT)	Wild type strain	Gillum, A. M., et. al.; Mol Gen Genet. 1984; 198:179-82.
<i>QDR1KO</i>	qdr1Δ::FRT/qdr1Δ::FRT	Dhamgaye, S., et. al.; Antimicrob Agents Chemother. 2012; 56:495-506
<i>QDR2KO</i>	qdr2Δ::FRT/qdr2Δ::FRT	
<i>QDR3KO</i>	qdr3Δ::FRT/qdr3Δ::FRT	
<i>QDR123KO</i>	qdr1Δ::FRT/qdr1Δ::FRT, qdr2Δ::FRT/qdr2Δ::FRT, qdr3Δ::FRT/qdr3	

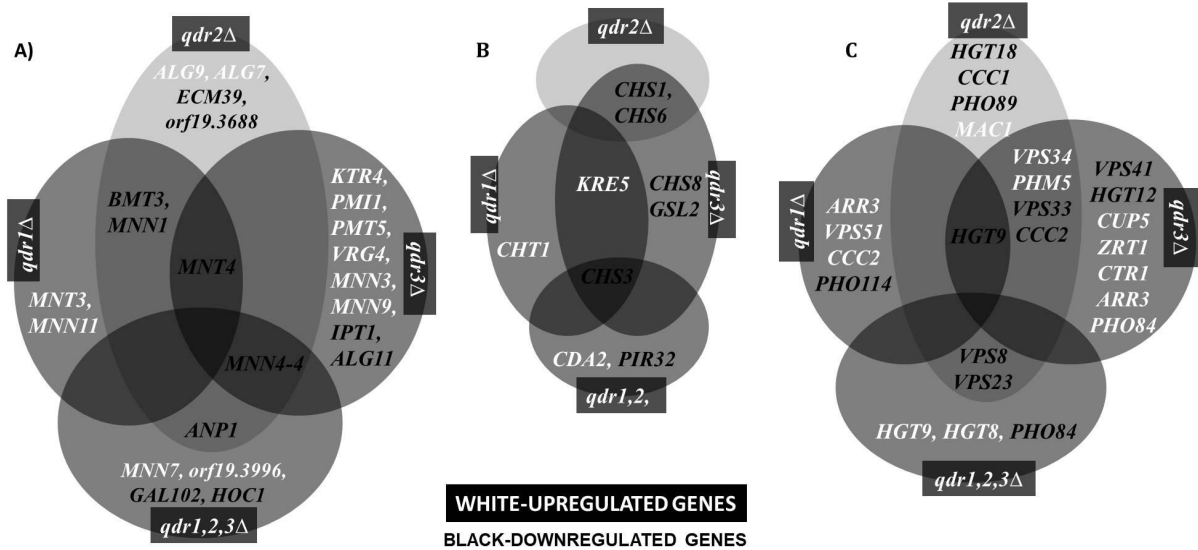
760 **Supplementary Table S2: Primers used in the study**

S.No	Gene Name	Primer Sequences (5'-3')
Primers used for RT-PCR analysis		
1.	ACT1	5'-TCGCTTTGGACTTCGAACAA-3'-Forward 5'-AGATGGAGCCAAAGCGGTAA-3'-Reverse
2.	HGT8	5'-TCTTTGTGGGTGCCATGTTA-3'-Forward 5'-ACCACCAGCCCATGTAGAAG-3'-Reverse
3.	HGT9	5'-GGGTGGCTGTTGTCTTCATT-3'-Forward 5'-GCAAATCCACCAAACCAACT-3'-Reverse
4.	MNN7	5'-GCCAGAATTGAACCAGGAGA-3'-Forward 5'-TCCCATTAGGGGCATCATAC-3'-Reverse
5.	VPS33	5'-ACTTACACCCGTTGGACGAC-3'-Forward 5'-CACTTCGAGTCACACCTCCA-3'-Reverse
6.	CHS6	5'-GGTCGAACAGCGTCAGAAAT-3'-Forward 5'-GATGTCCCCGTCATGTATCC-3'-Reverse
7.	CCC2	5'-TAATCGTCTTGGAGGCAACC-3'-Forward 5'-CAGCTGCAACTGGTGGTAAA-3'-Reverse
Primers used for qRT-PCR analysis		

8	HGT9F	TGATTGTCACCTGTTCCACC
9	HGT9R	CAAATGGTTCAGGTTTTGCC
10	HGT8F	GTTACCGGTATAAGGCTGGG
11	HGT8R	GCCAAAAAGCCTATGTCTGC

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Figure 1:



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Figure 2

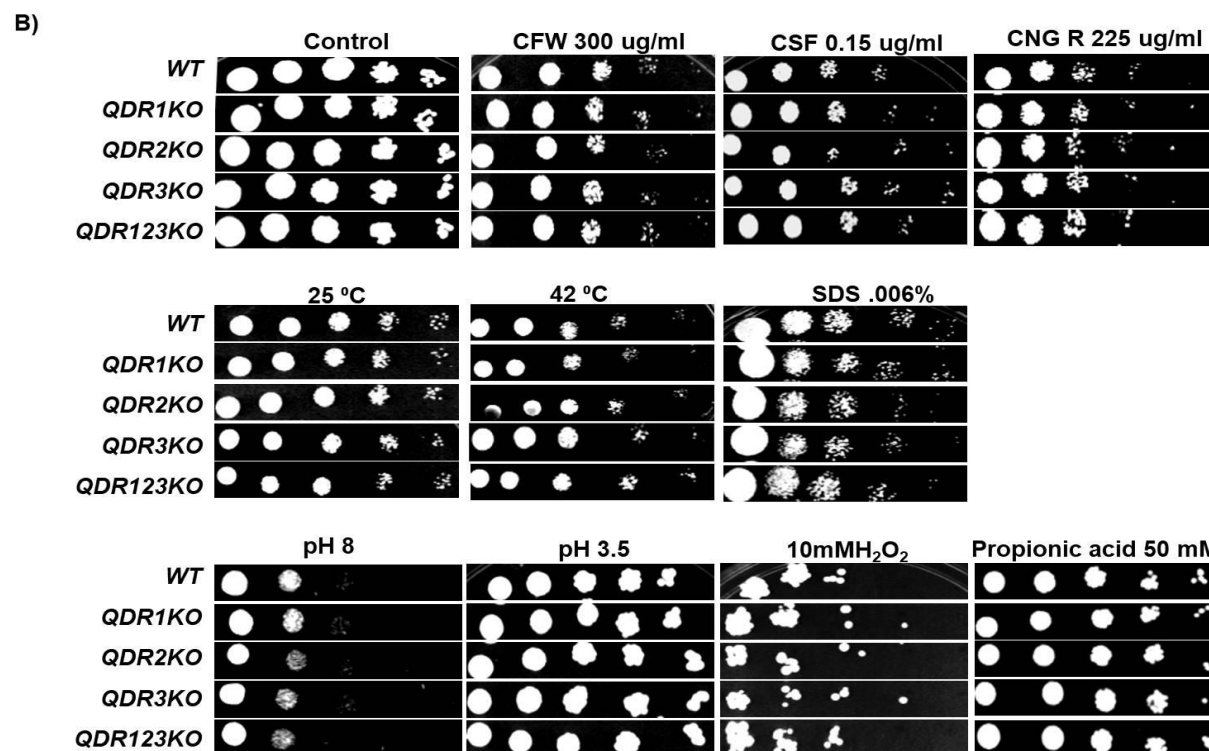
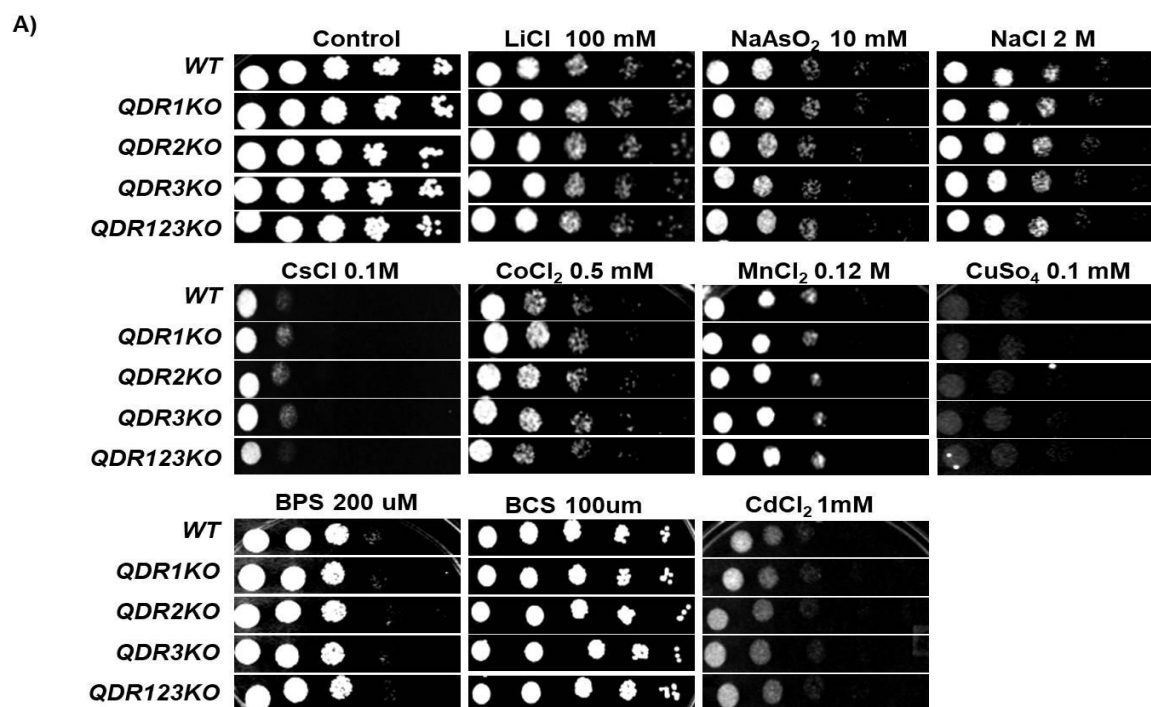


Figure 3

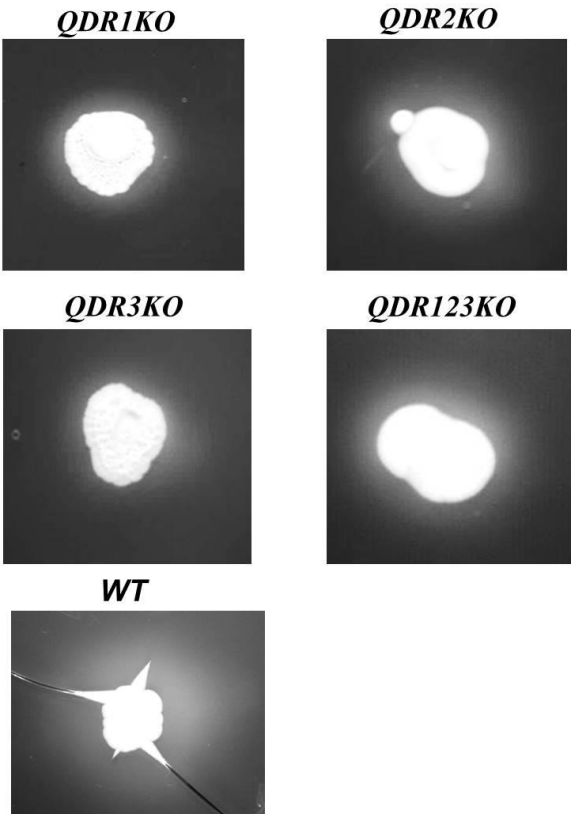


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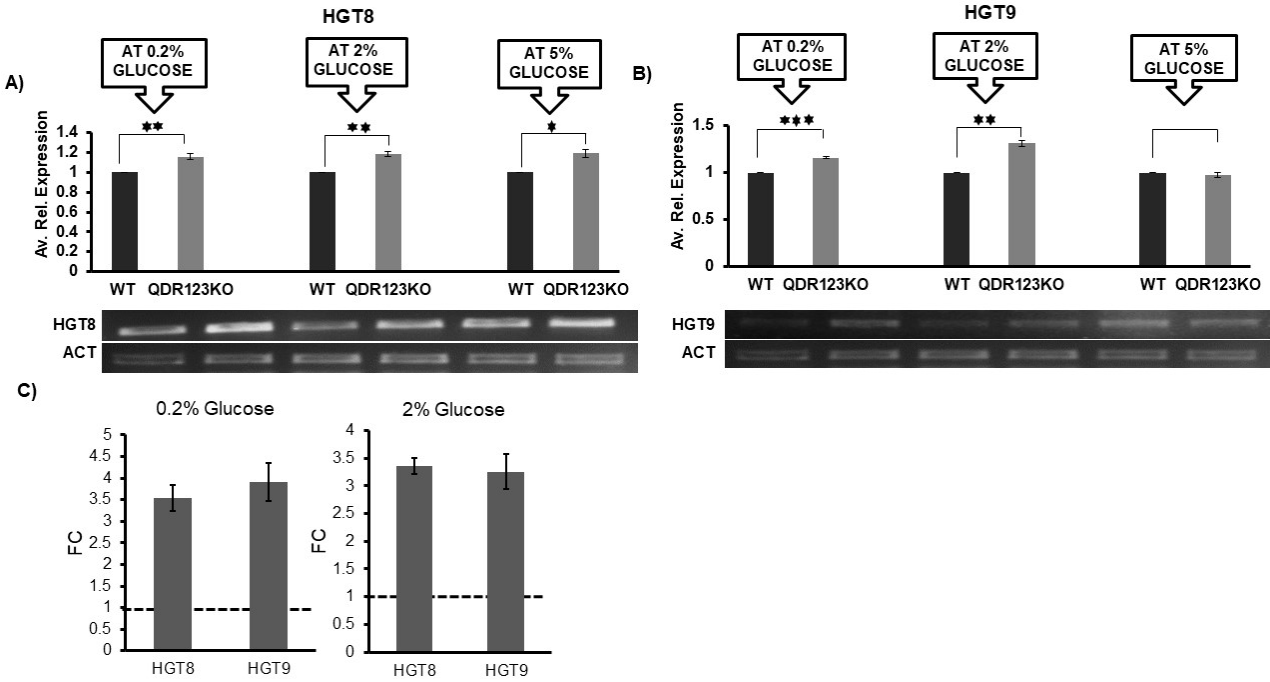


Figure 5

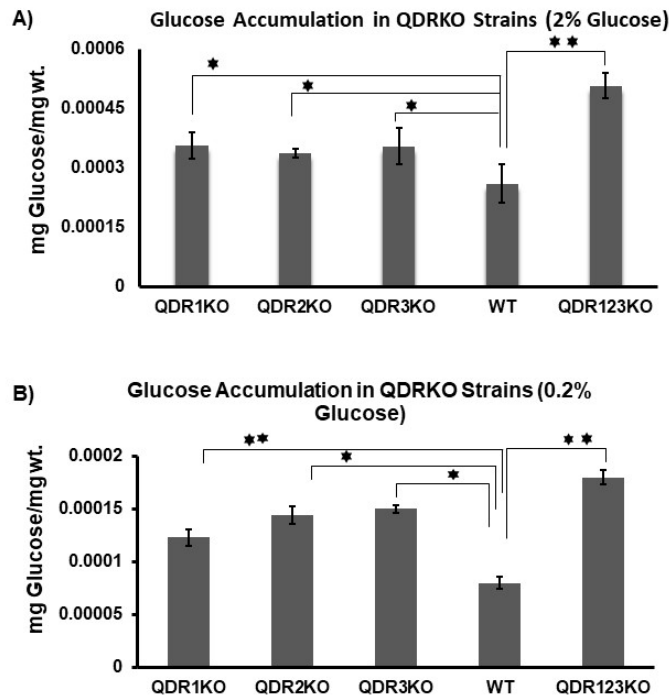
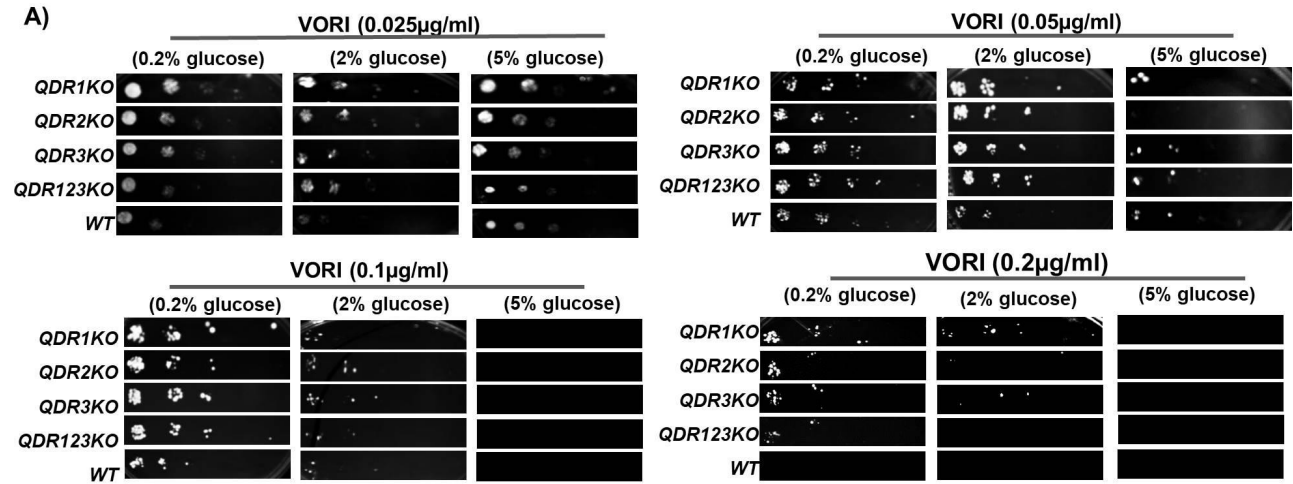


Figure 6



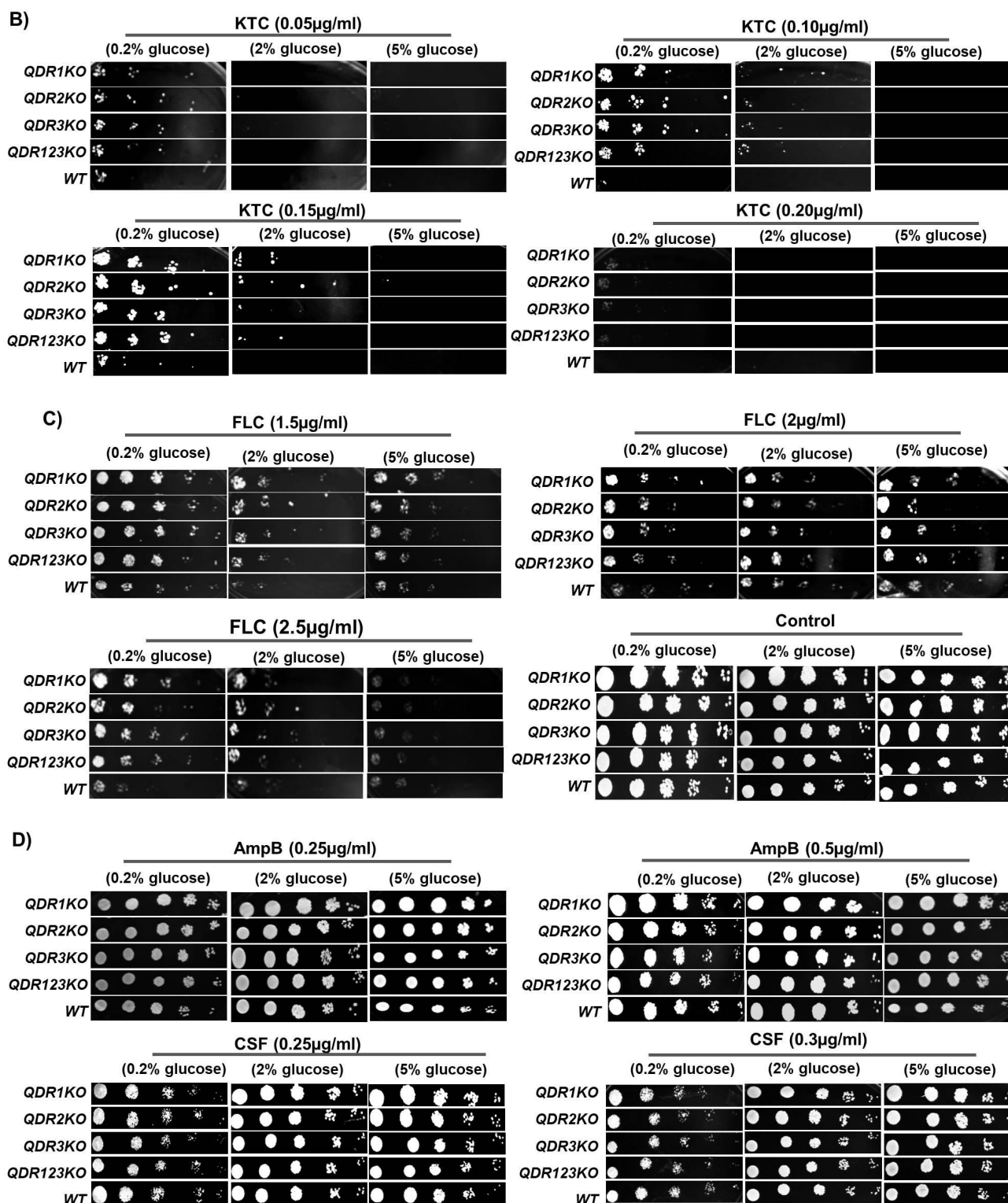
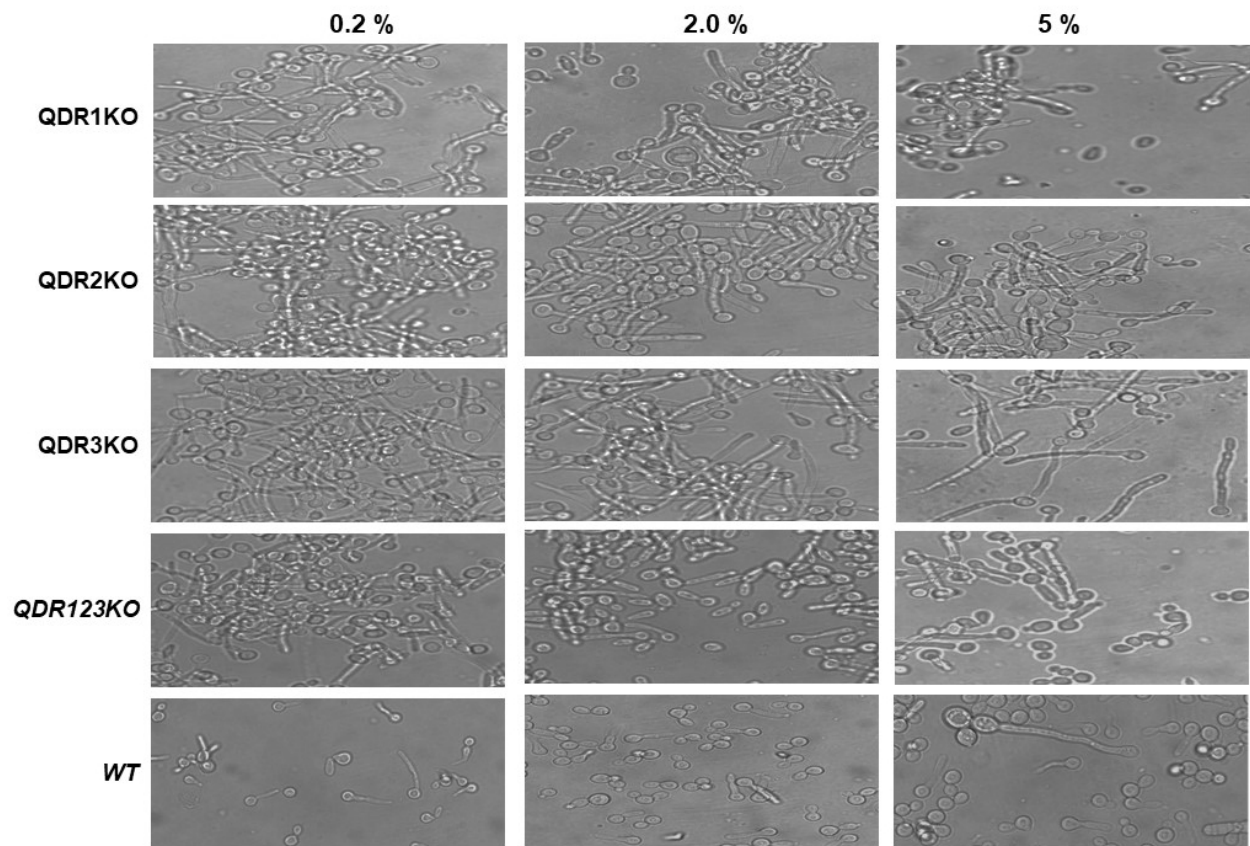


Figure 7



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Figure 8

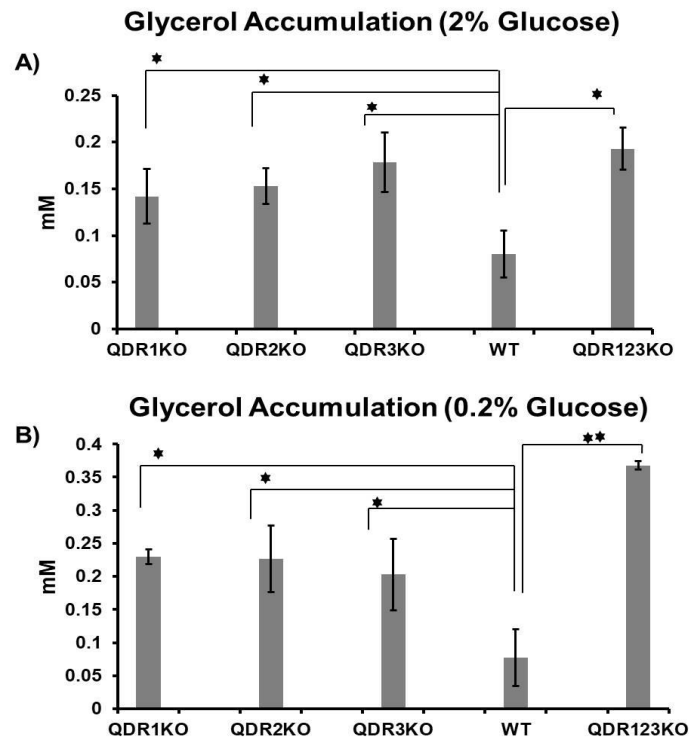
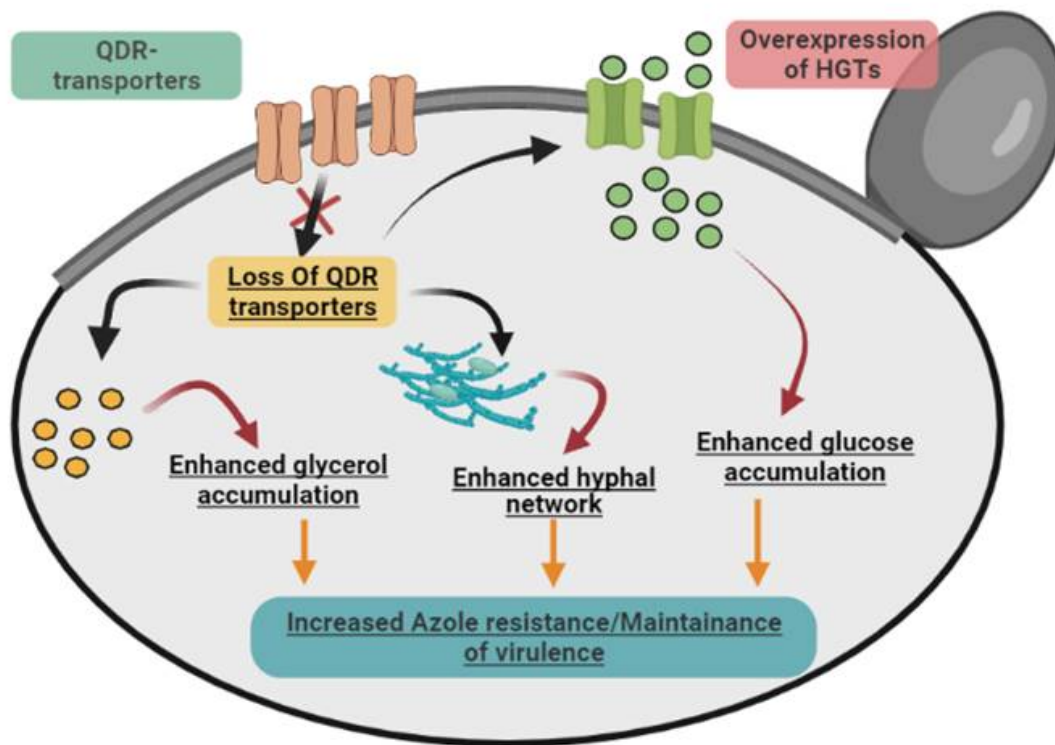
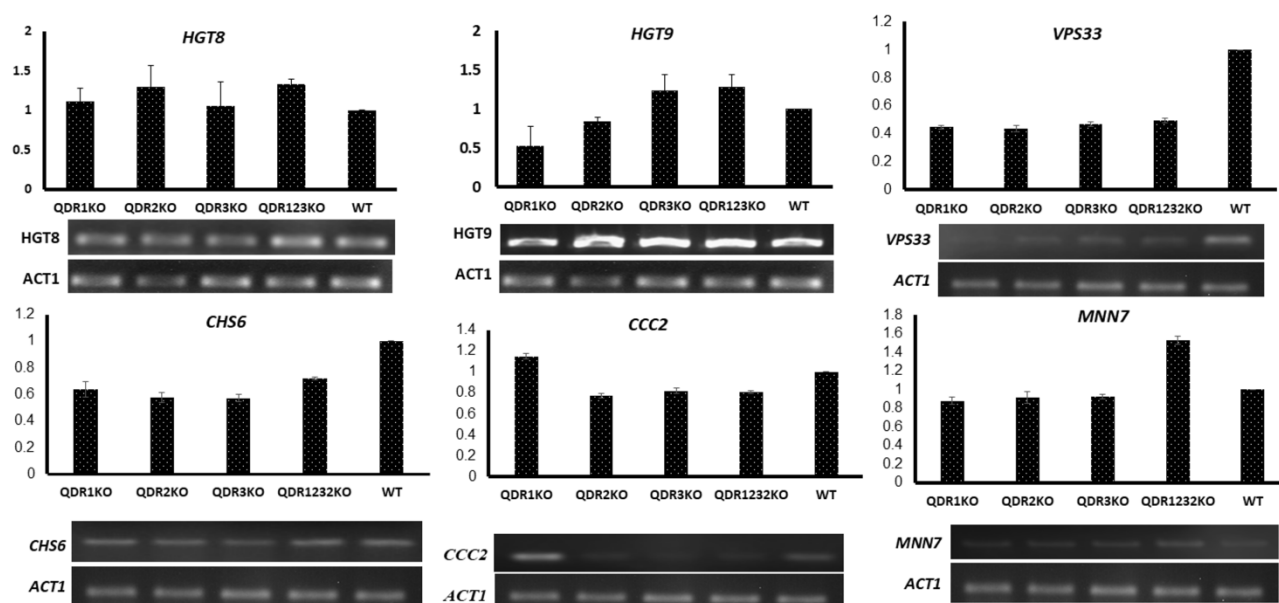


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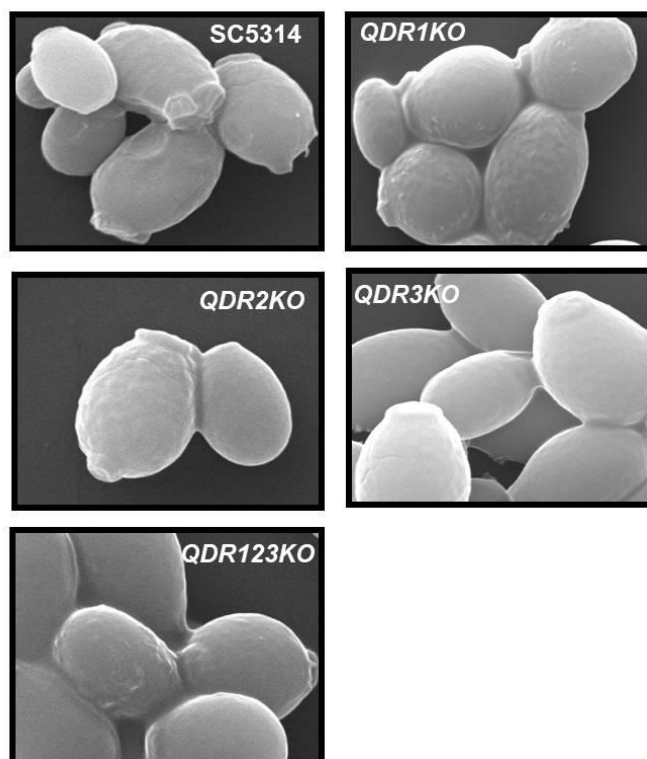


Supp. Figure 1:



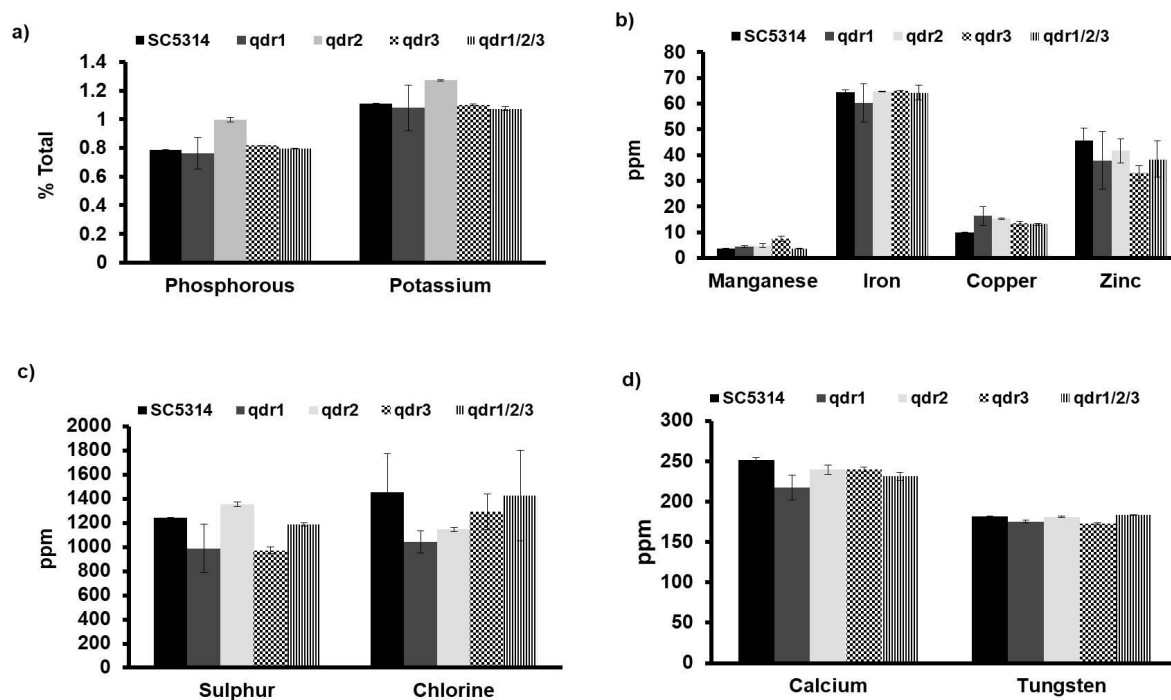
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Supp. Figure 2



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Supp. Figure 3



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